

A Incoherent Feedforward Loop (IFFL)

$$\frac{dx}{dt} = \alpha u - \delta x$$

$$\frac{dy}{dt} = \beta u - \gamma x y$$

Steady state:

$$x_{ss} = \frac{\alpha u}{\delta}, \quad y_{ss} = \frac{\beta \delta}{\gamma \alpha} \quad (\text{independent of } u)$$